UNIVERSITY OF UYO
DEPARTMENT OF MATHEMATICS
MTH 111 TEST FIRST SEMESTER 2020/2021 SESSION

Name of Student:_____

Reg. No._____ **GROUP**_____ **SERIAL No.**_____

Department:_____ **Signature**_____

PHYSICAL SCIENCES

Fill the spaces above first. Then answer ALL the Questions by writing the correct answer in the space provided. No phones, no calculators are allowed. Time allowed: 40 minutes

1. Let A = $\{x : 2x^2 + 3x - 2 = 0\}$ *and* $B = \{x: x^2 + 3x - 4 = 0\}$ find $(A \cup B) \times (A \cap B)$

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2. Find the values of p and q given that $x^4 + x^3 - 7x^2 + px + q$ is divisible by $(x-1)$ and $(x+2)$

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3. Find, without tables, the square root of $11 + 6\sqrt{2}$.

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4. Find the fourth term in the expansion of $(a - b)^8$

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5. Solve the equation for x, $\log_6(5x+1) = \log_{\sqrt{6}}(5x-1)$

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6. Solve $|2x - 1| < ⅓$

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7. Solve the equation $2^{2x}.27^x = 216$.

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8. Express as a sum of partial fractions $\dfrac{3x^2+8}{(x-2)(x^2+1)}$

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9. Let α, β be the roots of the equation $3x^2 + 5x + 2 = 0$, find the value of $\alpha^2 + \beta^2$

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10. For what values of λ does the equation $(\lambda - 2)x^2 + 4x + \lambda + 1 = 0$ have distinct real roots?